of each transition within the segment. In this paper, we only focus on segments with deterministic transitions, although the regret preference model equally accommodates segments with stochastic transitions. For a single deterministic transition (s_t, a_t, s_{t+1}), $regret_{\mathrm{d}}(\sigma_t|\tilde{r}) \triangleq V^*_{\tilde{r}}(s^\sigma_t) - [\tilde{r}_t + V^*_{\tilde{r}}(s^\sigma_{t+1})]$. For a full segment,

$$\begin{aligned} regret_d(\sigma|\tilde{r}) &\triangleq \sum_{t=0}^{|\sigma|-1} regret_d(\sigma_t|\tilde{r}) \\ &= V^*_{\tilde{r}}(s^\sigma_0) - (\Sigma_\sigma \tilde{r} + V^*_{\tilde{r}}(s^\sigma_{|\sigma|})), \end{aligned} \tag{3}$$

with the right-hand expression arising from cancelling out intermediate state values. Equation 3 only applies to deterministic transitions; Knox et al. (2022) provide the general formulation of regret. The default version of the regret preference model is the Boltzmann distribution over the *negated* regret:

$$P_{regret}(\sigma_1 \succ \sigma_2|\tilde{r}) = logistic\Big(regret_d(\sigma_2|\tilde{r}) - regret_d(\sigma_1|\tilde{r})\Big).$$

See Appendix B for further explanation of regret, of partial return, and of how these preference models differ. We refer to the regret and partial return of a segment as types of *segment statistics*.

Knox et al. (2022) demonstrated that the regret preference model better fits a dataset of human preferences, leading to more human-aligned learned reward functions. Knox et al. (2024) showed that the predominant method for fine-tuning large language models by RLHF can be derived from either preference model, but the derivation from the regret preference model avoids arbitrarily setting the discount factor $\gamma = 0$. From the regret preference model, Hejna et al. (2023) derive constrastive preference learning (CPL), which learns a policy *directly* from human preferences and thereby avoids the challenge during learning of modeling optimal value functions for $\hat{r}$ each time it is updated.

Nonetheless, using the partial return preference model remains appealing, given the greater simplicity of learning a reward function with it and the amount of research that has been built upon it. Therefore our goal is to assess whether it is possible to influence humans to adopt a given preference model, whether regret or partial return.

In the descriptions above of the two preference models, both are defined narrowly to assume that humans follow the Boltzmann distribution. This assumption is ubiquitous in RLHF but has limitations, which Zhu et al. (2024) overview. Both partial return and regret can, however, inform probability distributions other than the Bolzmann distribution. Therefore, in our evaluations (Section 5) we additionally test whether humans conform to the *noiseless* version of a preference model, which deterministically prefers the segment with the more desirable statistic (e.g., higher partial return for the noiseless partial return preference model or lower regret for the noiseless regret preference model).

4 Experimental task and preference elicitation procedure

To empirically investigate our methodology for influencing human preferences, we collected preference datasets labeled by human subjects with IRB approval. This section provides an overview of the user interface elements shared by each experiment introduced in Section 5. See Appendix C for further details.

Subject training Subjects first learn about the general grid-world delivery domain, for which task instantiations are created by a map of objects and terrain. The subjects specifically are taught about the reward function and how objects and terrain affect the agent's transitions. As part of this teaching, subjects control the agent on domain maps designed to teach one or two concepts at a time. One such map—the one used to collect preferences—is shown in Figure 3.

Figure 3: The delivery task shown to human subjects for gathering preferences. The yellow vehicle is the agent, and its objective is to maximize its score. Score maximization requires reaching the red inverted teardrop.

Preference elicitation interface After teaching subjects to understand the domain and delivery task, we elicit their preferences. Figure 4 illustrates a baseline version of the preference elicitation interface. In this work, we exclude preferences labeled "can't tell." After preference elicitation, a survey tests the subject's task comprehension and attentiveness, as detailed in Appendix C.4.

Figure 4: The baseline preference elicitation interface shown to humans annotators. All three of our experiments involve changes to this interface, whether by adding additional metrics (PRIVILEGED experiment), training the human before eliciting preferences (TRAINED experiment), or changing the elicitation question (QUESTION experiment).

5 Experimental evaluation of three methods of influence

Aiming to decrease the gap between an RLHF algorithm's assumed preference model and how a preference dataset is actually generated by humans, we conduct three random-assignment experiments for collecting human preferences. Each experiment represents a type of intervention and has three conditions that each result in a preference dataset: a control condition, an intervention condition that influences subjects to follow the regret preference model (P_{regret}), and an intervention condition that influences subjects to follow the partial return preference model (P_{Σ_r}). Thus, the ability of each type of intervention to influence the human towards a preference model is tested with two different preference models.

- **PRIVILEGED experiment** (Section 5.1) - The intervention of this experiment is to display information about each segment's regret or partial return under the ground-truth reward function during preference elicitation. We refer to this information as privileged because it relies upon the ground-truth reward function, which in practical settings is unknown by the code underlying the preference elicitation interface.
- **TRAINED experiment** (Section 5.2) - The intervention of this experiment is to train subjects to give preferences based upon either partial return or regret. This training procedure includes teaching people to calculate a segment's regret or partial return.
- **QUESTION experiment** (Section 5.3) - The intervention of this experiment changes the question during preference elicitation (see Figure 4, "Which path do you prefer?") to one designed to encourage adherence to one of the preference models.

In all experiments, the ground-truth reward function remains the same and is taught to subjects before preference elicitation to enable our analysis, but it is unavailable to the reward learning algorithm, which relies solely on the generated preference dataset. Only the PRIVILEGED experiment leverages the ground-truth reward function in its preference elicitation interface design.

5.1 PRIVILEGED experiment

We first study whether providing subjects with privileged information about a preference model during preference elicitation influences their preferences towards that model. Presenting this privileged information serves as a probe into how susceptible to influence human preferences are.

When presented with two segments for preference labeling, subjects in all three conditions are asked "Which shows better behavior?". This question was later refined for the other experiments to the baseline question in Figure 4. Each subject labeled preferences between 35 to 50 segment pairs. After data filtering (see Appendix C), our datasets come from from 64 subjects in the P_{Σ_r}-PRIVILEGED condition (video walk-through of the interface), 65 subjects in the P_{regret}-PRIVILEGED condition (video walk-through), and 50 subjects in the PRIVILEGED-Control condition (video walk-through). We refer to each condition's resultant preference dataset by the condition's name.

Intervention details: subject training In the P_{Σ_r}-PRIVILEGED condition, subjects are shown each segment's partial return during preference elicitation. Likewise, subjects in the P_{regret}-PRIVILEGED condition are shown each segment's regret. Examples can be found in Figure 16 of the Appendix. Note that we do not explicitly instruct subjects to use the displayed segment statistics when labeling preferences. Rather, it is made visible. In the third condition, PRIVILEGED-Control, subjects are not shown any information during preference elicitation other than the visualization of the two segments that all subjects in these experiments see. See Appendix D for further details. Conditions differ only as discussed above.

Hypothesis 1: Presenting each segment's statistic—whether partial return or regret—will influence the human to give preferences according to this statistic.

To test this hypothesis, we compare how well the target preference model—i.e., the preference model the intervention aims to influence human preferences towards—predicts the resultant dataset of preferences for the corresponding condition and for the control condition. Specifically, the preference model is given the ground-truth reward function, scaled by a constant positive factor. Given a target preference model, we compare the mean cross entropy (i.e., negative log likelihood) of each condition's preference dataset. Noting that all positive scalings of the reward function order policies equivalently and also only affect the Boltzmann distribution as a temperature parameter would, we choose the scaling constant that results in the lowest cross entropy during a grid search.

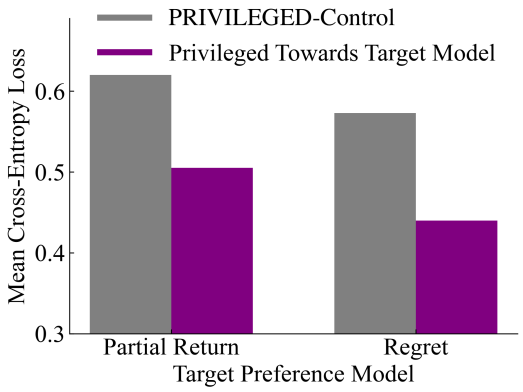

Figure 5: For the PRIVILEGED experiment, the mean cross-entropy loss over each condition's preference dataset with respect to the target preference model. If the loss is lower for an intervention's dataset than for the PRIVILEGED-Control dataset, then the former is better predicted by the target preference model. Performing a Mann-Whitney U test results in $p < 0.01$ for both conditions.

The cross entropy losses plotted in Figure 5 support Hypothesis 1. In particular, for both intervention conditions, presenting PRIVILEGED information about the target preference model results in a dataset with a higher likelihood than that of the PRIVILEGED-Control dataset. Because samples in each condition are unpaired, we perform a Mann-Whitney U test comparing the likelihood of samples in the PRIVILEGED-Control dataset to those in the intervention condition's dataset. Both interventions result in a statistically significant effect ($p < 0.01$). Appendix H provides plots of the cross-entropy loss for all reward scaling parameters for each experiment, and more details on the statistical test. A concern may arise that our proposed interventions merely teach human subjects more about the ground-truth reward function rather than about the target preference model. However, as we discuss in Appendix H, our full empirical results illustrate that this concern is unwarranted.

We remove the assumption that humans give preferences according to the Boltzmann distribution—used in the above likelihood test—by computing the accuracy for a noiseless version of each preference model. A Fisher's exact test also finds significant effects for both interventions ($p < 0.01$), supporting Hypothesis 1. See Appendix I for details about our accuracy testing.

Hypothesis 2: Presenting each segment's statistic leads to learning more aligned reward functions with the preference model corresponding to this statistic.

For each condition's dataset, we learn a reward function $\hat{r}$ by minimizing Equation 1 when assuming P_{regret} or P_{Σ_r}. In all experiments, regret is approximated via the algorithm proposed by Knox et al. (2022), employing successor features to efficiently compute differentiable estimates of the value functions in Equation 3. Value iteration (Sutton & Barto, 2018) then computes the approximately optimal $Q^*_{\hat{r}}$ function, and we then derive the maximum-entropy optimal policy from $Q^*_{\hat{r}}$, which uniformly randomly selects among all optimal actions. We assess the mean return of the resulting policy relative to the ground-truth reward function r over the initial state distribution D_0.

We normalize the mean return of a policy π, V^π_r, using the formula $(V^\pi_r - V^U_r)/V^*_r$. Note that V^*_r denotes the expected return of an optimal policy and V^U_r denotes the expected return of a uniformly random policy, both computed over D_0. A normalized mean return greater than 0 is better than V^U_r, while a value of 1 corresponds to optimal performance. We consider a normalized return above 0.9 to signify near-optimal performance.

For each dataset, we randomly assign human preferences to different numbers of same-sized partitions. Each partition is used as a training set, and we evaluate with 10 different random seeds that control how preferences are partitioned.